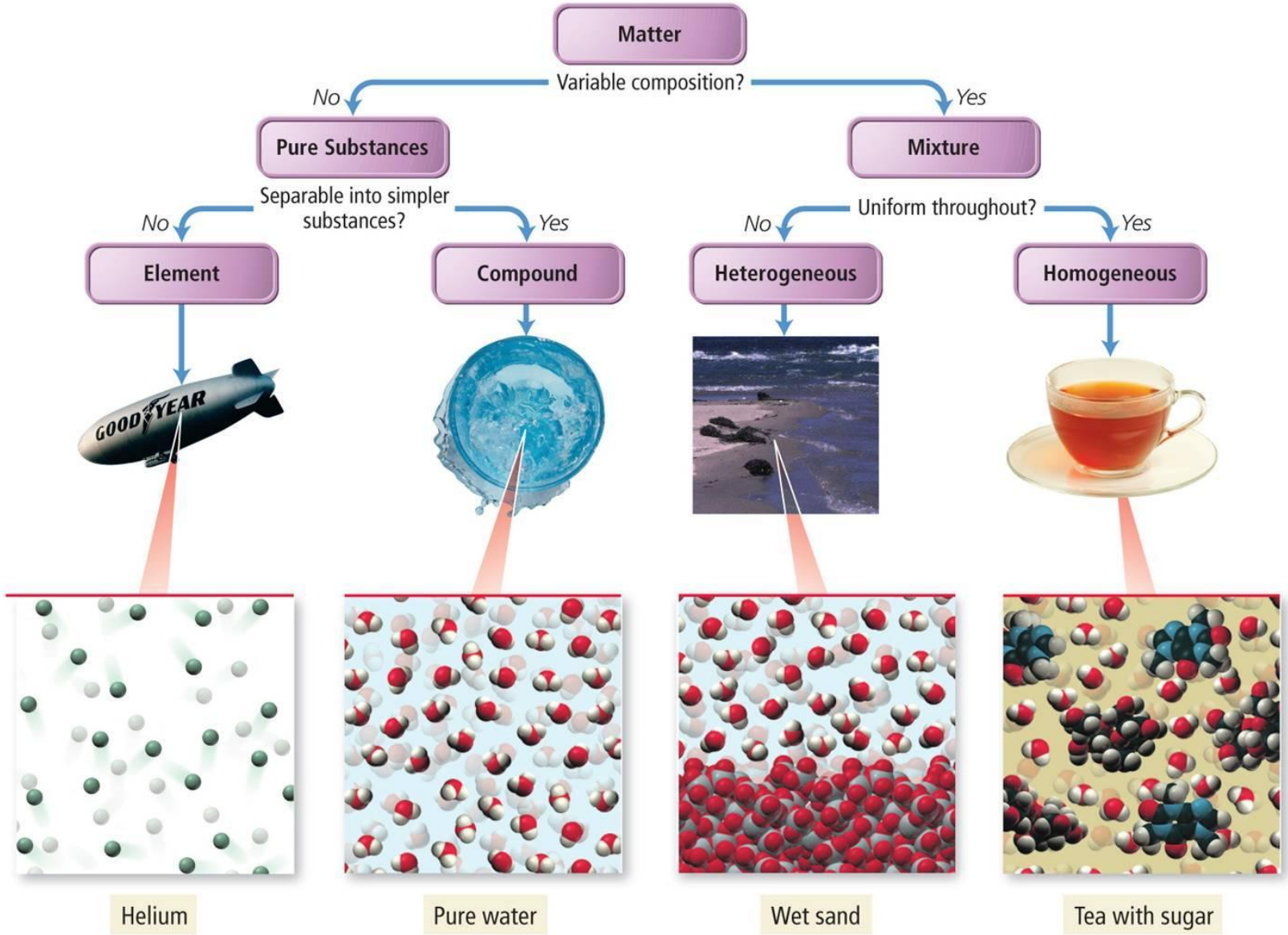
The Classification of Matter

Intro Activity

1) Get a lined piece of paper and number it 1-10
2) Get out your properties vocab sheet
3) With your group circulate around the room observing the samples and write a **minimum of 3 physical properties** for each sample. Do <u>NOT</u> open the containers.

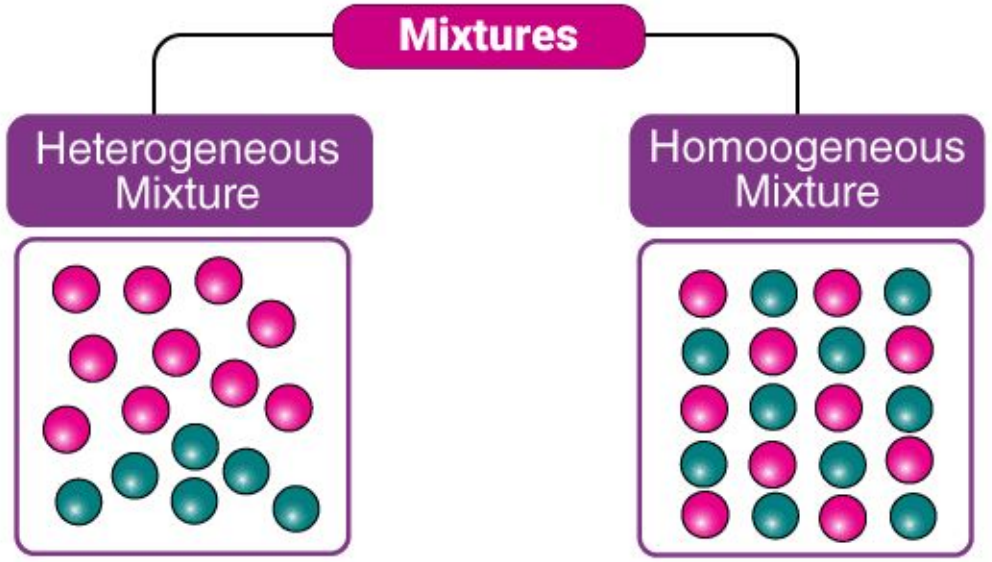
Classification of Matter



© 2010 Pearson Education, Inc.

Mixtures

- A substance that is made up of at least two **different** types of particles that are **physically** combined
- Classified based on particle **size**



Heterogeneous Mixtures

- **Large** particles that are completely **mixed** and can be **seen** with the naked eye or with a microscope
- Sometimes called a **mechanical** mixture

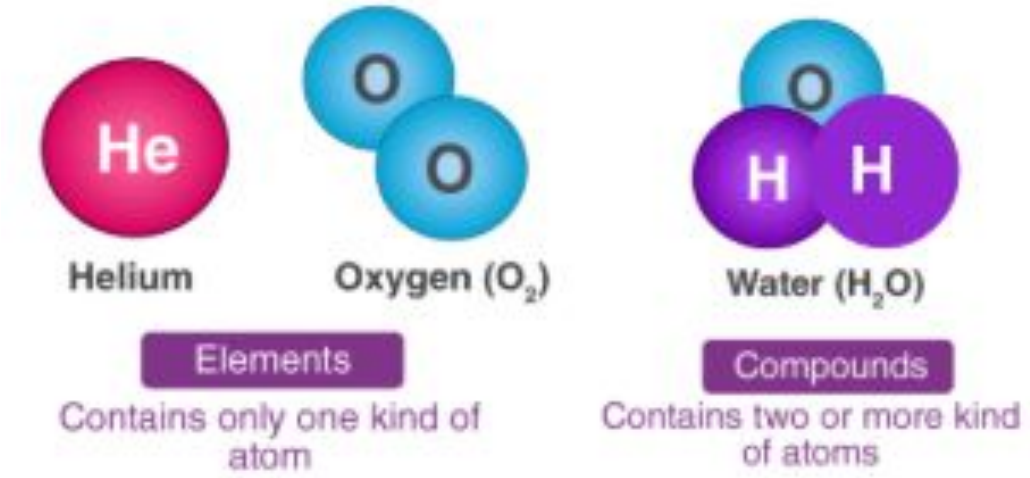
Homogeneous Mixture

- Contains very **small** particles mixed thoroughly together
- Appears the **same** throughout
- E.g. a **solution**

TEDEd
WHAT'S IN A MIXTURE?

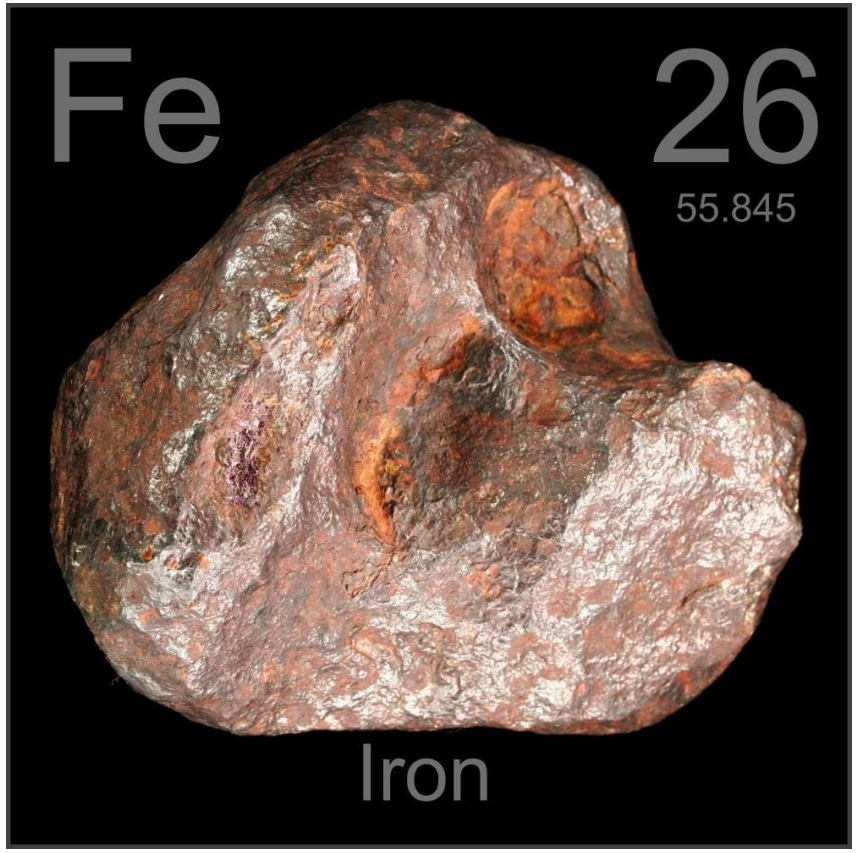
Pure Substances

- Composed of only **one** type of particle
- All samples have the same composition and **properties**
- Pure substances can **physically** combine to form **mixtures**



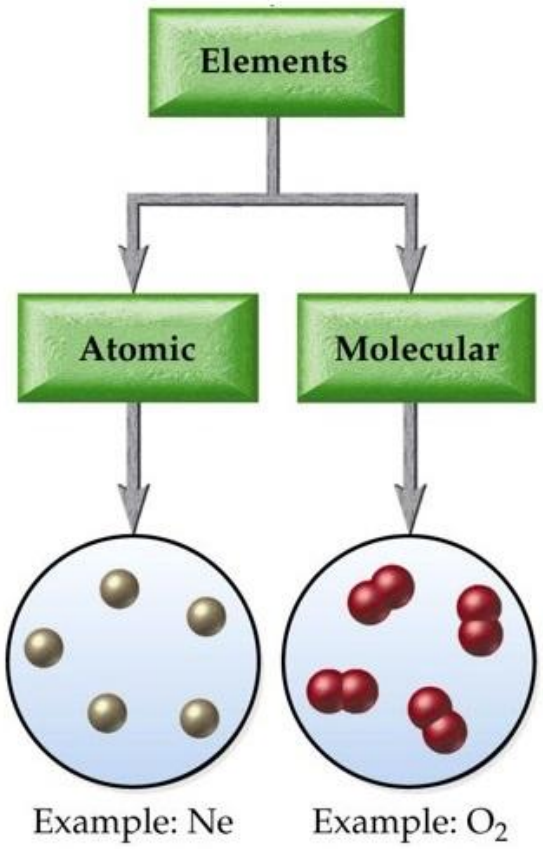
Element

- **Cannot** be **broken** down into simpler substances by physical *or* chemical means
- Made of unique particles called **atoms**



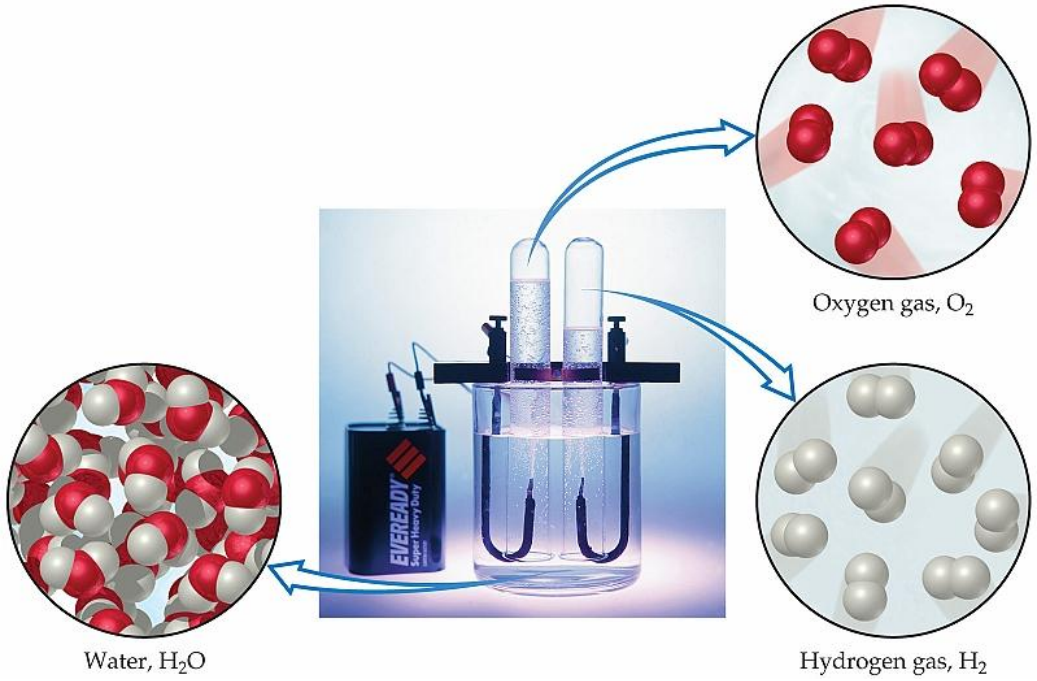
Element

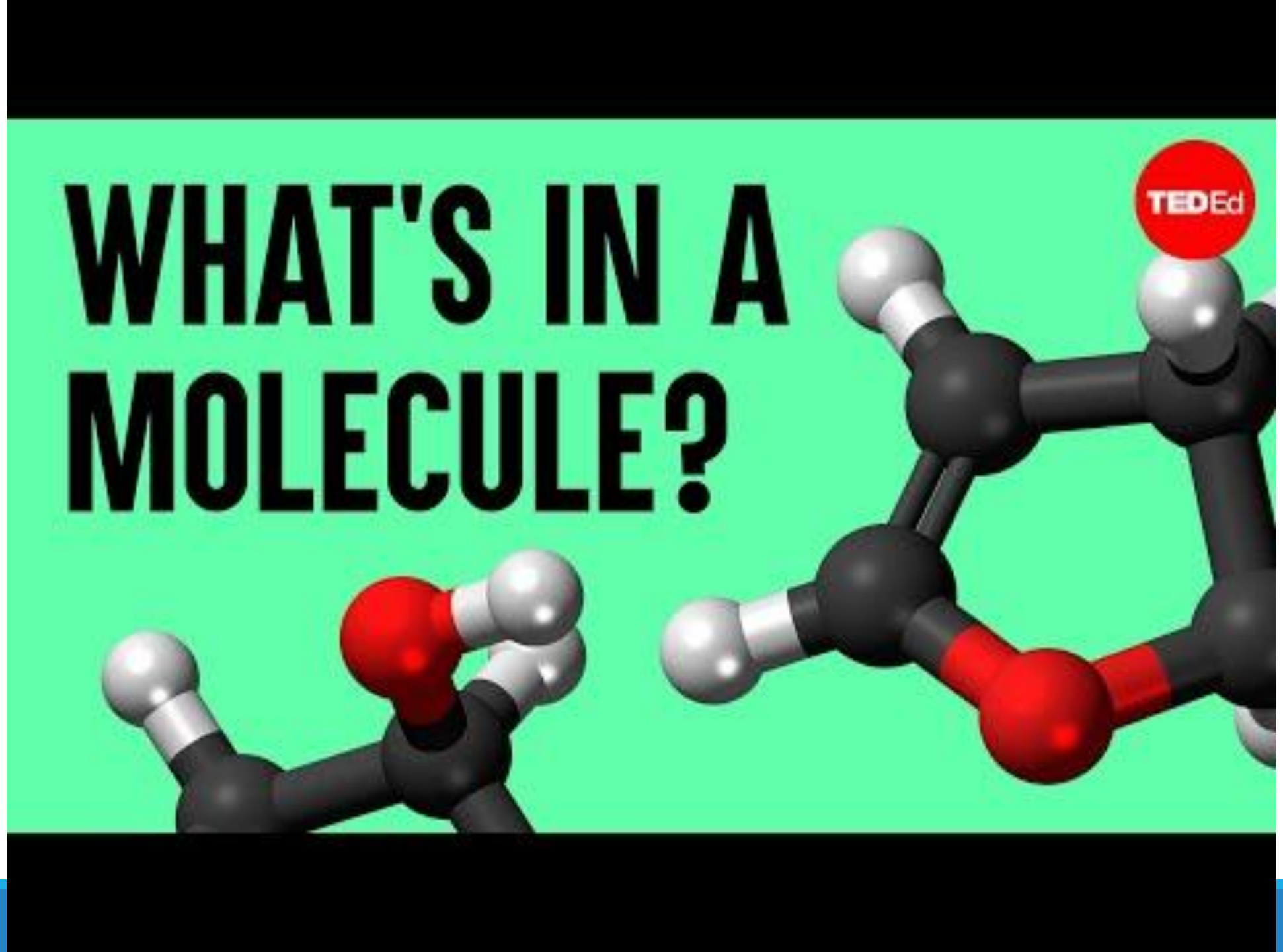
- Some elements can exist as **single** atoms, but several exist in **pairs** or in larger groups
- These are often referred to as **molecular** elements
- Keep in mind - there is still only **one** type of particle!



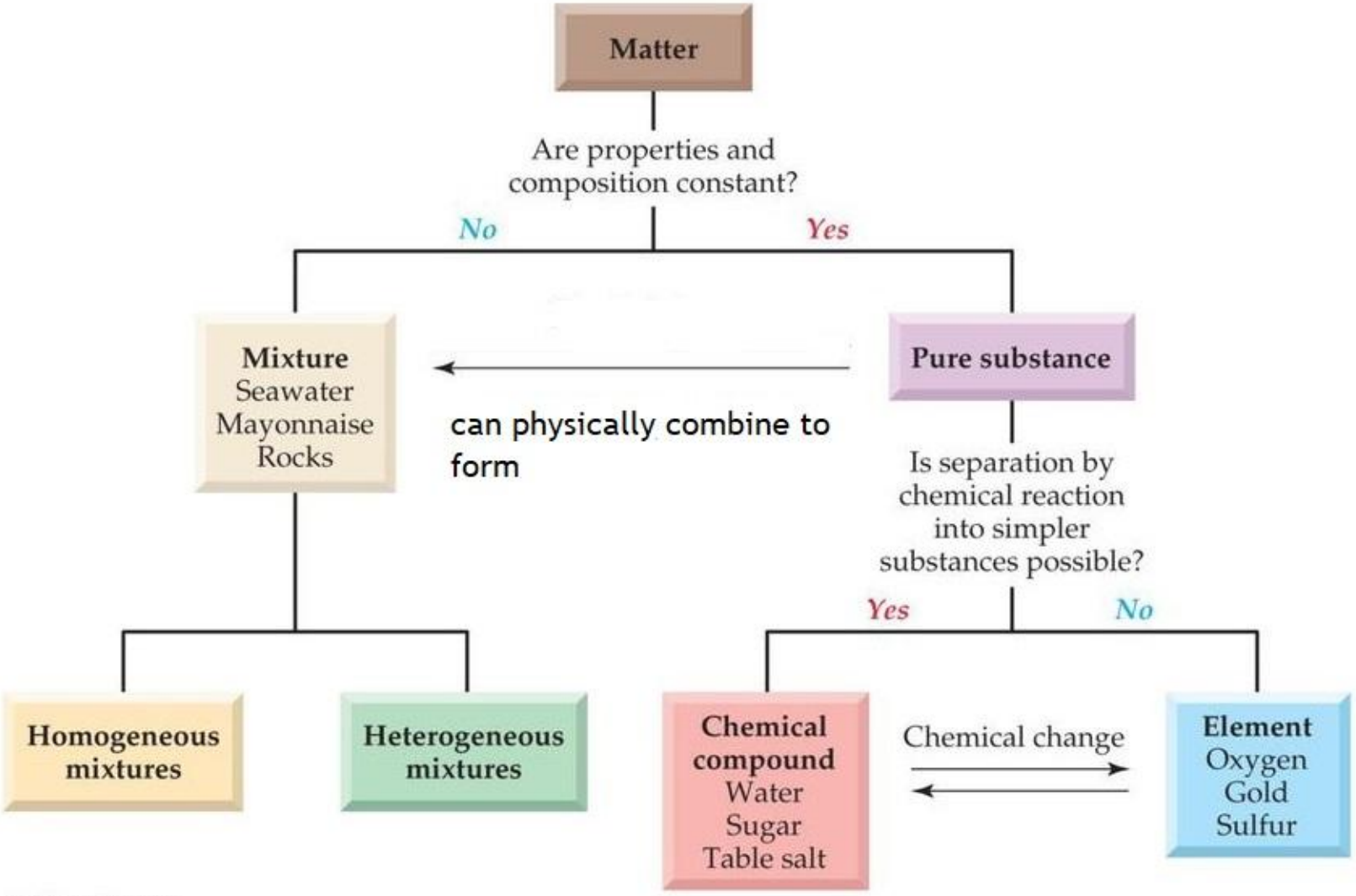
Compound

- Composed of two or more elements that are **chemically** combined together
- The elements are combined in **definite** proportions
- Can be broken down into elements again by **chemical** means (requires energy)




WHAT'S IN A
MOLECULE?
TEDEd

Summary Chart



© 2013 Pearson Education, Inc.

Your turn

- Classifying Matter Practice